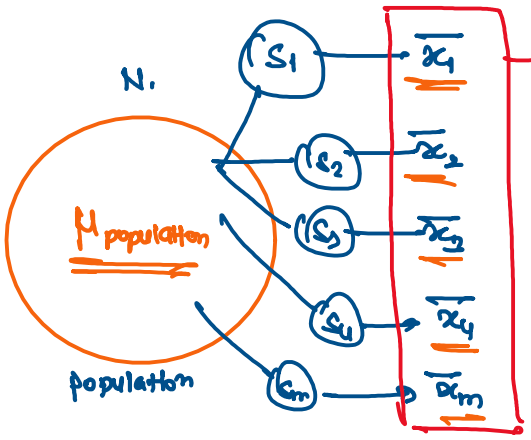
Tuesday, 23 December 2025 6:02 PM

- We have covered uptill now.
  - Central limit theorem.
    - → It is a fundamental concept of statistics.
    - → that describes the behaviour of the sampling distribution of the sample mean.



→ Sampling distribution of sample means. ($\bar{x}$).

$$\frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \dots + \bar{x}_m}{m} = \mu_{\bar{x}}$$

- There are usually 3 conclusions.

According to CLT, the sampling distribution of sample mean $\bar{x}$

1. will be approximately normal distribution, regardless of the shape of the original population distribution

$$\bar{x} \sim N(\mu_{\bar{x}}, \sigma_{\bar{x}})$$

2. will have a mean this is approximately equal to the population mean

$$\boxed{\mu_{\bar{x}} \approx \mu_{population}}$$

3. will have a standard deviation that is equal to the population standard deviation divided by the square root of the sample size.

$$\boxed{\sigma_{\bar{x}} \approx \frac{\sigma_{population}}{\sqrt{n}}}$$

- To approximate the point estimate we have one of the biggest challenge that is data collection problem.

sample ~~sam~~ size = 50.
no. of sample = 1000.
→ 50,000 ← no. of data point that we need to collect

- Uptill now.

→ can be bias

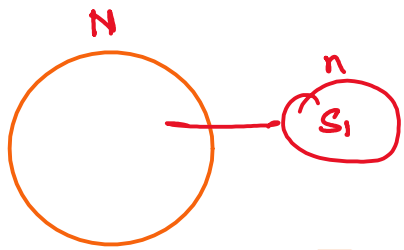
Point estimate using 1 sample $\boxed{\mu_{sal} \approx \bar{x}_1}$ not a good approximation to make inference

- Point estimate using m samples & CLT.

$\boxed{\mu_{sal} \approx \mu_{\bar{x}}}$ → less bias.
→ very good approximation
→ data collection is st

(N) — (m)
↳ each of sample size n.

- Better option to make estimation → Confidence Interval estimate using a single data point & CLT.

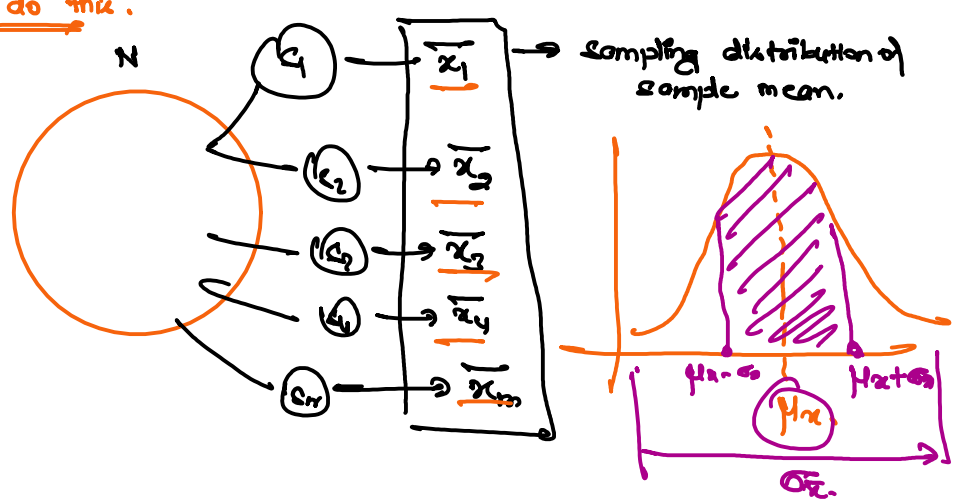


$\boxed{\mu_{population} \approx \bar{x}_1}$ → mean of a single sample.

This estimate has alot of bias, in order to cater the bias.

$\boxed{\mu_{population} \approx \bar{x}_1 \pm \text{something}}$ → with this range estimate, we can also associate some confidence

- How to do this.



$\bar{x}_1$ → a part of normal distribution

$\boxed{\mu_{population} \approx \bar{x}_1 \pm 1 \times \sigma_{\bar{x}}}$ → confidence level 68% —(i)

Similarly

$\mu_{pop} \approx \bar{x}_1 \pm 2\sigma_{\bar{x}}$ → the confidence interval will be 95%.

Similarly, $\mu_{pop} \approx \bar{x}_1 \pm 3\sigma_{\bar{x}}$ → the confidence interval

of 44.4 %.

Now according CLT.

$$\boxed{\sigma_{\bar{x}} \approx \frac{\sigma_{pop}}{\sqrt{n}}}$$

Can I rewrite the first eqn.

$$\boxed{\mu_{pop} \approx \bar{x}_1 \pm 1 \times \frac{\sigma_{pop}}{\sqrt{n}}} \rightarrow 68\% \text{ confidence interval.}$$

To make the above formula general

$$\boxed{\mu_{population} \approx \bar{x}_1 \pm z_{\alpha/2} \times \frac{\sigma_{pop}}{\sqrt{n}}}$$

$\bar{x}_1$ → point estimate of a single sample

$\frac{\sigma_{pop}}{\sqrt{n}}$ → standard error.

$z_{\alpha/2}$ → with y% confidence.

confidence interval estimate. →

$$\boxed{\bar{x}_1 \pm z_{\alpha/2} \times \frac{\sigma_{pop}}{\sqrt{n}}} \rightarrow \text{interval.}$$